

Econometric Games Preliminary: Estimating Causal Effect of STEM

Student Share on Median Incomes

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ABSTRACT

We estimate the effect of a college's STEM major percentage on its graduates' median earnings using a two-stage least squares framework using college-level longitudinal and cross sectional data of incomes and other characteristics of the student body. We account for selection bias through institution-specific fixed effects, and find after instrumental variable robustness checks STEM-heavy colleges have modest but statistically significant advantages in median earnings, consistent with the findings of other literature.

1. Introduction

A significant economic literature examines how different college majors lead to varying future earnings outcomes. One common finding is that choosing majors in Science, Technology, Engineering, and Mathematics (STEM) or in Business and Economics leads to higher earnings compared to choosing a major in the liberal arts. Many studies suggest that earnings differences across majors are comparable in magnitude to the earnings gap between high school and college graduates. As a result, major choice can strongly affect labor market outcomes and plays an important role in students' educational decisions.

Using data from the National Longitudinal Study of the Class of 1972 (NLS72), Arcidiacono (2004) studies major choice and earnings outcomes in 1986. He finds that students in the natural sciences (including engineering and mathematics) earn the highest wages, followed by students in Business (including Economics), who significantly outperform those in the social sciences and humanities. The researcher uses a dynamic three-period model in which students initially choose a major in period 1, are allowed to change majors in period 2, and enter the labor market in period 3. This framework helps mitigate selection bias related to ability, as students with a weaker level may switch to easier majors. Arcidiacono finds that students with the highest math ability, measured by SAT scores, tend to sort into the natural sciences. Similarly, Kirkebøen et al. (2016) study Norwegian students and find that choosing science over the humanities nearly triples early-career earnings. Andrews et al. (2024) follow public K–12 students in Texas through college and into the labor market to study the growth and variability of earnings across and within majors. They find that higher-return majors also exhibit lower earnings variability, making them more attractive to students. In addition, they show that high school and college fixed effects play an important role in shaping these estimates, a result that is particularly relevant for our analysis.

While these studies focus on individual students and their characteristics, Chetty et al. (2020) adopt a different perspective by examining intergenerational mobility through college-level characteristics. They find that when students from different socioeconomic backgrounds – measured by parental income – attend the same college, earnings differences are substantially reduced. However, they also show that students from wealthier backgrounds are more likely to attend high-earning colleges, which limits intergenerational mobility. They therefore advocate for changes in admission policies to promote greater access. In our research, we also adopt a college-level perspective by focusing on class characteristics, studying the proportion of STEM students as a potential determinant of earnings outcomes across institutions. We use parental income, gender, and race as control variables to isolate the role of a college’s STEM representation in shaping future earnings.

We develop a framework that exploits time-series data to estimate college fixed effects. In a two-stage regression process, in which we incorporate information from changes in earnings for a given college over time, we capture persistent institutional characteristics, such as student culture, academic ambition, or labor market reputation. If omitted, these factors could generate substantial selection bias in our estimates. We also corroborate our analysis by implementing a split-sample instrumental variable approach developed by Angrist and Krueger (1995), applied to the returns to schooling, to ensure the robustness of our model in estimating the STEM effect, while also taking advantage of the time-series dimension of the data and capturing college fixed effects. We consistently find that, after controlling for student body and college characteristics, a higher proportion of STEM students within a school significantly increases students’ earnings outcomes. Across both estimation strategies, a 1% increase in the share of STEM students leads to an increase of more than 120 dollars in earnings for the concerned institution. Our approach thus complements the existing student-level evidence on the earnings returns to STEM majors by emphasizing the role of college-level characteristics in shaping long-run labor market outcomes.

2. Data

We use data from Opportunity Insights, presented in Chetty et al. (2020), to obtain longitudinal data for 1,543 institutions¹ identified with an Office of Postsecondary Education Identification number (OPEID). The panel consists of the dependent variable of interest, median incomes, as well as time-varying controls such as male/female share, share of married, and median parental income for 12 years. Median incomes are measured by grouping by the birth year of the graduating class, from 1980 to 1991, and looking at those students' median income in the year 2014. The regressor of interest, STEM Share, is only available in the year 2000, alongside other important control variables like the Barron ranking of the institution, and racial decompositions for this year. In addition to lacking data, these factors tend to be time invariant, making a panel data model to directly estimate the effect of STEM Share unfeasible.

Summary statistics of the data are provided in the Appendix in Tables 1 and 2.

3. Econometric Framework

Following the limitations of our data, we employ both panel and only cross-sectional techniques, suppressing the subscript t where necessary. Let median earnings for students of college i in graduating class t be represented by y_{it} , and x_i be percentage of students in STEM. Let further w_i be the $k \times 1$ vector of control variables² outlined above, and π a $k \times 1$ vector of coefficients.

We specify a simple linear relationship between the dependent variable, regressor of interest, and control variables. The naïve model to estimate the effect of STEM percentage for college i is thus

$$y_i = \beta_0 + \beta_1 x_i + w_i^\top \pi + u_i \quad (1)$$

¹ Around 500 institutions were removed from the dataset due to missing values for key variables.

² Here, $\top$ denotes the transpose of a vector

However, this cross-sectional model omits the unobserved variables leading to selection bias.

3.1. Selection Bias and Controls

An important aspect of any regression of wages on education factors is selection bias, i.e. the presence of omitted or unobservable variables that drive individuals income and correlate with education. Heckman et al. (2006) established that cognitive and noncognitive skills explain a variety of labor market outcomes. Moreover, Webber (2014) found that accounting for selection bias significantly alters earnings premia associated with different college majors, the object of this study.

In our case, students at a certain college could be more ambitious or have a greater cognitive ability, which would lead to those students obtaining higher income. Therefore, we assume that each college admits an institution-specific effect α_i that represents latent qualities of the student body at college i affecting the earnings of their graduating classes. We are unable to directly measure these factors through the covariates available in our dataset, so we instead turn to a panel data model decompose the error from (1) into

$$u_{it} = \alpha_i + \lambda_t + v_{it} \quad (2)$$

where λ_t is a time-specific fixed effect to reflect the general upward trend of median incomes across graduates of all colleges.

3.2. Two Stage Regression and Small Sample Bias

To estimate the model absent of selection bias, we make use of the available time-series observations and first obtain estimates for the fixed effects and then augment the cross sectional regression of median income on STEM percentage with the fixed effects as a synthetic control variable.

Literature that seeks to estimate particular values for these unobservable factors for controls or other regressions often specifies random effects, where in the decomposition (2) α_i is correlated with the idiosyncratic error v_{it} . Under this regime, fixed effects estimation yields noisy estimates and biased standard errors, as noted in Murphy and Topel (2002). Both Kane and Staiger (2008) and Chetty et al. (2014) recover a teacher-specific valued added term in their regression of test scores on students and use a shrinkage estimator with Bayesian priors on the distribution of α_i to counter the bias introduced by noise. However, for our purposes the α_i are best modeled as fixed effects, since random effects assumes the effects are random draws from an underlying distribution where knowing the the name or ID of teacher doesn't give any information. Colleges are different, since we use an exhaustive list of institutions, and in many cases the name or brand of an institution does provide significant information about its student body's ability. It is reasonable then that our estimates do not suffer from these issues, and moreover only serve as a control for the relationship between the variable of interests.

Nonetheless, given the modest size of our panel the estimates of college fixed effects could suffer from measurement error, which could cause attenuation bias in the second-stage regression. As a robustness check, we therefore follow a similar approach to Angrist and Krueger (1995), using a split-sample instrumental variable approach to estimate the fixed effects. The idea is to split the panel dataset in half, and obtain separate estimates of the fixed effects using each half (say, $\{\hat{\alpha}_i^1\}_{i=1}^N$ and $\{\hat{\alpha}_i^2\}_{i=1}^N$ respectively), and then use $\{\hat{\alpha}_i^2\}_{i=1}^N$ as instruments for $\{\hat{\alpha}_i^1\}_{i=1}^N$. Now, for every i , $\hat{\alpha}_i^1$ and $\hat{\alpha}_i^2$ are both estimates of α_i , the true intrinsic characteristics of the college, so $\hat{\alpha}_i^2$ is a relevant instrument for $\hat{\alpha}_i^1$. At the same time, $\hat{\alpha}_i^2$ is uncorrelated with the measurement error of $\hat{\alpha}_i^1$, making it a valid instrument, under the assumption that the error associated with the first half of the sample is uncorrelated with the error associated with the second half of the sample. This assumption seems innocuous given that exogenous shocks to student cohorts (like a specific graduating class's

labour outcomes) likely depend on the graduating student cohort itself, which changes throughout the sample.

3.3. Estimating the Model

For the first stage, we follow the standard techniques for fixed effects estimation outlined in Stock and Watson (2011) and employ a $n - 1$ and $T - 1$ dummy variable technique to obtain estimates for institution specific effects, with the following regression equation

$$y_{it} = \beta_0 + \sum_{j=2}^N \gamma_j Dj_i + \sum_{s=2}^T \delta_s Bs_t + b_{it}^\top \theta + v_{it} \quad (3)$$

where Dj_i and Bs_t are dummy variables that take the value 1 when $i = j$ or $t = s$, and 0 otherwise, and v_{it} is the idiosyncratic error term. (3) also includes the vector of panel control variables b_{it} . We can then back out what each entity specific effect α_i is by the relationship $\alpha_1 = \beta_0$, $\alpha_i = \beta_0 + \gamma_i$ for $i \geq 2$. Therefore, we obtain via OLS

$$\hat{\alpha}_i = \hat{\beta}_0 + \hat{\gamma}_i \quad (4)$$

Finally, the estimated $\hat{\alpha}_i$ serves as a control in the second regression

$$y_i = \beta_0 \hat{\alpha}_i + \beta_1 x_i + w_i^\top \pi + \varepsilon_i \quad (5)$$

where ε_i satisfies the control variable assumption that $E(\varepsilon_i | \hat{\alpha}_i, x_i) = E(\varepsilon_i | \hat{\alpha}_i)$, allowing $\hat{\beta}$ to be estimated via OLS unbiasedly.

To perform the split-sample instrumental variable robustness check, we follow a four-step procedure:

1. We first obtain estimates $\{\hat{\alpha}_i^1\}_{i=1}^N$ of $\{\alpha_i\}_{i=1}^N$ using regression equation (3) with the truncated dataset containing data for the first 6 years, birth years 1980-1985.
2. We similarly obtain estimates $\{\hat{\alpha}_i^2\}_{i=1}^N$ of $\{\alpha_i\}_{i=1}^N$ using the truncated dataset containing data for the last 6 years, birth years 1986-1991.

3. Using $\{\hat{\alpha}_i^2\}_{i=1}^N$ as instruments for $\{\hat{\alpha}_i^1\}_{i=1}^N$, we perform the regression

$$\hat{\alpha}_i^1 = \chi_0 + \chi_1 \hat{\alpha}_i^2 + v_i \quad (6)$$

obtaining estimators $\hat{\chi}_0$ and $\hat{\chi}_1$ which we use to obtain the fitted values $\tilde{\alpha}_i = \hat{\chi}_0 + \hat{\chi}_1 \hat{\alpha}_i^2$.

4. We perform the cross-sectional regression specified in (5), including $\{\tilde{\alpha}_i\}_{i=1}^N$ as the fixed effect terms.

We next evaluate the results obtained using this methodology, comparing the base case that does not account for fixed effects, the case that follows the standard Stock and Watson (2011) approach, and the case inspired by Angrist and Krueger (1995). Note that we report HC3 heteroskedasticity-robust standard errors which do not account for our use of generated regressors throughout our process. While a correction is technically warranted, due to time constraints we abstract from this problem in this paper, taking comfort in the fact that the t-statistic for our primary regressor (STEM share) is >10 using naive standard errors, so a correction would likely not alter the statistical significance of our results.

4. Results

The results of the regression without fixed effects are summarized in Table 4, and the regression including fixed effects is in Table 5 in the Appendix. We find that while the STEM share ordinarily has a significant and positive effect, the inclusion of fixed effects reduces this almost half, from 216.386 to 128.118, while continuing to be highly significant. Interpreting this coefficient, a student deciding to go to a college with a 1% higher STEM share can expect a \$128 increase in their income.

Similarly, the coefficient on the median parent income control variable also reduces in magnitude from 0.191 to 0.027 and loses its significance from the 1% to the 5% level, as if controlling for

the college characteristics, outcomes for students' earnings depended less on their socioeconomic background. This result is consistent with Chetty et al. (2020) findings of lower disparities in income once students attended the same college. The effect of sticker price approximates 0, first being significant at the 10% level without controlling for college fixed effects before turning negative in the second regression and gaining in significance to the 1% level.

Both regressions exhibit surprising results when coming to race, as in Table 4 it can be seen that the model estimates that a 1% share of Asian students or Hispanic students within an institution would lead to almost a 25,000 dollars and 8,000 increase in earnings, respectively. Meanwhile, the coefficient on the Black share of students is negative and insignificant. When accounting for college fixed effects, however, Asian and Hispanic shares lose their significance (with even a change in signs for the Hispanic variable) while the coefficient on the shares of Black students increases in magnitude and becomes significant at the 1% level. In both regressions the Alien coefficient is positive but insignificant. The dummies representing college ranking and prestige remain large and significant at the 1% level, but also decrease in magnitude as additional unobserved characteristics of colleges are included within the model.

The split-sample instrumental variable results, displayed in Table 6 in the Appendix, largely reaffirm the results of the first two regression processes, demonstrating the robustness of our findings. Indeed, STEM share continues to have a significant and positive effect with a coefficient of 193.484, which is in the same realm as in the other regressions. As expected, median parent income continues to exhibit a positive effect on median graduate earnings, as do the Barron's Rank tier variables. The effect of racial composition continues to be noisy, with results that are not robust across our regression methodologies. Interestingly, the decrease in the R^2 to 0.800 compared to 0.905 in the non-instrumental-variable fixed effects case is consistent with the removal of spurious explanatory power stemming from measurement error; yet because the coefficient of STEM share

remained similar regardless of the methodology, we have confidence in this causal effect.

5. Conclusion

College major has an important impact on future earnings outcomes. Moving beyond the effect of major choice on the individual student, we were able to rigorously examine institutional-level outcomes, explicitly accounting for selection bias by measuring fixed effects through a two-stage least squares framework. We found that the composition of majors has an important impact on the median earnings across a college's entire graduating class, STEM share having a statistically significant positive effect. Using a split-sample instrumental variables approach to estimating the fixed effects demonstrated that our results are robust.

Our work lends credence to heuristics regarding the dynamics of career outcomes post-graduation, in particular relating to network effects. Indeed, one interpretation of our results is that a higher proportion of STEM students on a particular campus has positive spillover effects on non-STEM majors, for example through increased attention to the college by major high-paying employers that seek out technically trained students; though to rigorously quantify this interpretation, it would be necessary to compare our institution-level results with individual student-level results, which we relegate to future study.

Ultimately, our findings of positive institutional-level effects of STEM share on median graduate earnings aligns with the prior literature on college major selection, shedding light on the role of students' cohort of peers on their future labor market outcomes.

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APPENDIX

TABLE 1: Summary Statistics (Cross-Section)

Statistic	N	Mean	St. Dev.	Min	Max
Median Earnings (k_median)	1,543	38,363.25	12,581.80	11,400	98,500
Median Parental Income (par_median)	1,543	81,400.45	29,161.52	23,900	226,700
STEM Share (pct_stem_2000)	1,539	14.77	13.86	0.00	100.00

TABLE 2: College Characteristics and Student Composition

Statistic	N	Mean	St. Dev.	Min	Max
Sticker Price (2013)	1,526	15,785.12	14,196.85	1,104.00	51,008.00
Enrollment (2013)	1,543	8,921.93	14,522.97	247	231,384
Asian/Pacific Share (2000)	1,541	0.04	0.06	0.00	0.72
Black Share (2000)	1,541	0.11	0.16	0.00	1.00
Hispanic Share (2000)	1,541	0.07	0.11	0.00	0.96
Graduation Rate (2013)	1,449	0.45	0.23	0.05	1.00

TABLE 3: OLS with Fixed Effects

	<i>Dependent variable:</i>
	Median Earnings (y_i)
female	−13,926.940*** (847.153)
k_married	40,998.760*** (498.016)
par_median	−0.055*** (0.004)
College FE	Yes
Cohort FE	Yes
Observations	18,476
R ²	0.936
Adjusted R ²	0.930
Residual Std. Error	3,297.628 (df = 16,919)
F Statistic	157.784*** (df = 1,556; 16,919)

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

TABLE 4: Determinants of College-Level Earnings

	<i>Dependent variable:</i>
	Median Earnings (y_i)
STEM Share (2000)	216.386*** (22.472)
Median Parent Income	0.191*** (0.014)
Sticker Price (2013)	0.040* (0.021)
Asian/Pacific Share (2000)	24,779.970*** (4,537.811)
Black Share (2000)	-537.705 (1,002.089)
Hispanic Share (2000)	8,256.274*** (1,391.549)
Alien Share (2000)	8,008.859 (9,887.296)
Elite (Barron's Rank 1)	12,846.730*** (1,883.213)
Highly Selective (Barron's Rank 2)	10,777.640*** (1,401.222)
Selective (Barron's Rank 3–5)	7,813.896*** (469.846)
Constant	12,527.690*** (1,038.573)
Observations	1,522
R ²	0.782
Adjusted R ²	0.781
F Statistic	542.660*** (df = 10; 1511)

Note: HC3 heteroskedasticity-robust standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

TABLE 5: Determinants of College-Level Earnings (Including Estimated College Effect)

	<i>Dependent variable:</i>
	Median Earnings (y_i)
STEM Share (2000)	128.118*** (12.546)
Median Parent Income	0.027** (0.012)
Sticker Price (2013)	−0.056*** (0.014)
Asian/Pacific Share (2000)	1,654.547 (2,205.090)
Black Share (2000)	−10,787.140*** (752.778)
Hispanic Share (2000)	−1,540.437 (1,018.069)
Alien Share (2000)	790.452 (5,466.208)
Estimated College Effect ($\hat{\alpha}_i$)	0.811*** (0.037)
Elite (Barron's Rank 1)	8,261.185*** (1,080.399)
Highly Selective (Barron's Rank 2)	5,071.210*** (759.602)
Selective (Barron's Rank 3–5)	4,422.369*** (289.316)
Constant	33,499.220*** (1,117.676)
Observations	1,522
R ²	0.906
Adjusted R ²	0.905

Note: HC3 heteroskedasticity-robust standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

TABLE 6: College-Level Earnings Regression with Instrumented Fixed Effects

	<i>Dependent variable:</i>
	Median Earnings (y_i)
STEM Share (2000)	193.484*** (20.077)
Median Parent Income	0.128*** (0.015)
Sticker Price (2013)	0.061*** (0.020)
Asian/Pacific Share (2000)	15,415.380*** (4,110.158)
Black Share (2000)	-1,915.235** (976.802)
Hispanic Share (2000)	4,130.025*** (1,427.823)
Alien Share (2000)	11,116.770 (9,044.474)
Instrumented College Effect (cleanFE)	0.345*** (0.042)
Elite (Barron's Rank 1)	10,943.670*** (1,735.397)
Highly Selective (Barron's Rank 2)	8,075.335*** (1,272.520)
Selective (Barron's Rank 3–5)	6,075.698*** (470.301)
Constant	20,397.610*** (1,355.565)
Observations	1,522
R ²	0.801
Adjusted R ²	0.800

Note: HC3 heteroskedasticity-robust standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The college fixed effect is instrumented using split-sample estimates.